Sadeep Madhushan

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Portfolio Sadeep-Madhushan SadeepMadhushan

Profile

First-Class Honours graduate in Electronic and Telecommunication Engineering from the University of Moratuwa, Sri Lanka, with robotics research experience at the Singapore University of Technology and Design (SUTD) and industry experience as a software engineer. Experienced in developing robotic systems end to end, from SolidWorks-based mechanical design and Gazebo simulation to ROS 2-based control, motion planning, and navigation on real hardware.

Research Interests: Reinforcement learning for robotics, learning-based manipulation and motion planning, sim-to-real transfer, human-robot interaction

Education

University of Moratuwa, Sri Lanka *Jun. 2021 – Jun. 2025*

B.Sc. (Hons) Electronic and Telecommunication Engineering

- Graduated with **First-Class Honours (GPA: 3.84/4.00)** and **4x Dean's List** (Semesters 1, 4, 6, 8)
- Relevant coursework: Robotics (kinematics, DH parameters), Autonomous Systems (Kalman filters, EKF, SLAM), Machine Vision, Deep Learning for Vision

Nalanda College, Colombo *Jan. 2017 – Dec. 2019*

G.C.E. Advanced Level Examination 2019 (Physical Science Stream)

- **Z-Score 2.4172** with **3 A passes** (Combined Mathematics, Physics, Chemistry)
- Island rank **246** out of 30,000 students (top 0.8%) and district rank **80** in Colombo

Research & Professional Experience

Robotics Research Intern – Singapore University of Technology and Design *Nov. 2023 – May 2024*

Industrial training supervised by Prof. Tan U-Xuan (verify)

- Contributed to an autonomous hospital-cleaning robot project, focusing on manipulator control for high-touch surface cleaning.
- Developed a Gazebo simulation to evaluate planned robotic arm motions against its joint constraints.
- Integrated ROS2 and MoveIt framework with the simulation for manipulator control, motion planning, and trajectory execution.
- Tested and validated planned motions on a Universal Robots UR10e industrial manipulator.

Software Engineer – DirectFN Pvt Ltd. *May 2026 – Present*

- Develop end-to-end software features using Java and Angular, covering data models, backend APIs, and frontend integration within a layered architecture.
- Optimise Oracle SQL queries and implement frontend/backend caching to improve performance.

Software Engineer – IFS R&D International Pvt Ltd. *Jul. 2025 – Apr. 2026*

- Refactored legacy C# code into loosely coupled, interface-driven extensions and performed root cause analysis to resolve technical issues.
- Developed an N8N workflow integrating a vector database and AI to automate unit test generation for migrated code.

Robotics Projects

Mobile Robot Receptionist: A Semi-Humanoid Service Robot (verify) *Aug. 2024 – May 2025*

(ROS2 | MoveIt | Gazebo | C++ | Python | SolidWorks)

- Designed a 4-DoF robotic arm in SolidWorks for autonomous elevator-button interaction.
- Integrated camera-based elevator-button detection using a machine learning model.
- Configured the MoveIt framework for inverse kinematics and motion planning.
- Developed a Gazebo simulation for testing and integrated the system with physical hardware.

Turbobot: Autonomous Mobile Robot (Self-Initiated) (verify) *Sep. 2024 – Jan. 2025*

(ROS2 | Nav2 | Gazebo | RViz | C++ | Python | SolidWorks)

- Designed a 4-wheel differential-drive robot from scratch and developed its 3D model in SolidWorks.
- Integrated the robot model with ROS 2 and tested autonomous navigation in Gazebo.
- Built the physical robot and integrated a Raspberry Pi with wheel encoders and IMU sensors for sensor fusion and robot-state estimation.

Flexobot: 6-DoF Robotic Arm (Self-Initiated) (verify) *Sep. 2024*

(ROS2 | MoveIt | Gazebo | RViz | C++ | SolidWorks)

- Designed and 3D-modelled a 6-DoF robotic arm using SolidWorks.
- Integrated the arm with ROS2 and MoveIt for inverse kinematics and trajectory planning.
- Simulated the robotic arm in Gazebo to validate motion planning and joint movements.
- Developed C++ ROS2 nodes for joystick-based independent control of individual joints.

Machine Learning & Vision Projects

Parasitic Egg Detection and Classification *Oct. 2023*

(Python | Image Processing | Machine Learning)

- Applied the RTM-Det-tiny model for single-stage object detection, optimizing accuracy by slicing larger images into patches.
- This involved training and performing inference on the sliced images, then recombining them for final results.

Ophthalmic Biomarker Detection *Aug. 2023*

(Python | PyTorch)

- Identifying the presence of structural biomarkers in eyes. Implementing supervised and self-supervised learning techniques to make use of labeled and unlabeled data.

Achievements

- **2nd Runners-Up, RoboRoarz 2024, Singapore** – reconfigurable robot navigation (verify)
- **Champions, Robot Design Competition 2023 (Virtual)** – Webots simulation, University of Moratuwa
- **Champions, SparkLink 1.0 (2023)** – HDL competition, University of Ruhuna (verify)

Technical Skills

Programming Languages: C++, Python, Java, C#
Robotics Frameworks: ROS2, MoveIt, Nav2
Simulation & Visualisation: Gazebo, RViz, Webots
CAD & EDA: SolidWorks, Altium, Multisim, Proteus
Tools: Git, Docker, N8N

Certifications

Udemy: Robotics and ROS 2 – Learn by Doing! Manipulators (verify)
Udemy: Robotics and ROS 2 – Learn by Doing! Odometry & Control (verify)
Coursera: Supervised Machine Learning: Regression & Classification (verify)
Coursera: Advanced Learning Algorithms (verify)
Coursera: Unsupervised Learning, Recommenders, Reinforcement Learning (verify)
Coursera: Introduction to Computer Vision and Image Processing (verify)

References

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Commendation letter